

A
Mini Project Report
on
“Digital Market Yard”

SUBMITTED TO THE PUNYASHLOK AHILYADEVI HOLKAR SOLAPUR
UNIVERSITY, SOLAPUR IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF

**BACHELOR OF TECHNOLOGY DEGREE IN
COMPUTER SCIENCE AND ENGINEERING**

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SOLAPUR - 413 255
2024-25

AFFILIATED TO



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DECLARATION

We hereby declare that the work embodied in this Project Mini Project Report Entitled “**Digital Market Yard**” is carried out by us in partial fulfillment of the degree Bachelor in Technology (Computer Science and Engineering) from N. B. Navale Sinhgad College of Engineering, Solapur and we have not submitted the same to any other University/Institute for the award of any other degree.

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ACKNOWLEDGEMENT

We feel profound happiness in forwarding this Mini Project Report as an image of sincere efforts. Our sincere thanks to respected **Prof. Harish T. Gurme**, our project guide, for providing healthy environment and facilities in the department. He allowed us to raise our concern and worked to solve it by extending his co-operation time to time.

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We give our special thanks to respected Head of Department Prof. H. T. Gurme for their constant interest and encouragement throughout the completion of our project.

Goal makes us to do work. Vision is more important than goal which makes us to do work in the best way to make work equally the best. Thanks to our respected Principal, Dr. Shankar. D. Nawale for his support and vision.

We express our deep gratitude to all staff members who lend me their valuable support and cooperation to enable me to complete my project successfully.

We also thankfull to all friends for their extended support and valuable guidance.

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ABSTRACT

The Digital Market Yard System is an innovative web-based platform designed to modernize the traditional agricultural marketing process by providing a unified digital interface for farmers, merchants, and brokers. The current manual system used in agricultural market yards is often inefficient, non-transparent, and prone to delays and financial discrepancies. This project aims to bridge that gap through digital transformation.

The system allows farmers to list their crops with detailed specifications, while merchants can place direct orders or engage in real-time bidding. Automated modules handle tax and commission calculations, generate invoices, and maintain transaction history with cloud-based storage for security and scalability.

Developed using Spring Boot, Java 17, Thymeleaf, and MySQL, the platform ensures robust performance, secure authentication via Spring Security, and efficient data management. Future enhancements include mobile app integration, real-time chat, multi-language support, and analytics dashboards to further empower stakeholders.

This project supports the broader goal of increasing transparency, reducing dependency on intermediaries, and enhancing the income and decision-making capabilities of farmers through technology.

Mini Project Outline

The Mini Project Report is organized chronologically in terms of the objectives accomplished.

- **Chapter One** gives information about Recommendation System.
- **Chapter Two** gives literature review on the relevant areas required for the current research which contains approaches considered for network performance.
- **Chapter Three** gives information about Existing Systems and SDLC Model
- **Chapter Four** figures of ER Diagram , DFD Level 0 , DFD Level 1, DFD Level 2
- **Chapter Five** UML Diagrams with Use Case Diagram ,Class Diagram ,Sequence Diagram
- **Chapter Six** REQUIREMENT SPECIFICATION with Hardware Requirements and Software Requirements
- **Chapter Seven** Technologies used for Implementation with module 1 , module2 , module3
- **Chapter Eight** CONCLUSION AND FUTURE SCOPE
- **A Appendix** lists of abbreviation used in the Mini Project Report.
- **B References** lists the bibliography for the system.

Contents

1	INTRODUCTION	1
1.1	Introduction to Project	1
1.2	Problem Statement	1
1.3	Scope and Objectives	2
1.3.1	Scope	2
1.3.2	Objective	2
1.4	Identification of Need	3
2	LITERATURE REVIEW	4
2.1	Literature Review	4
2.2	Existing Systems	4
2.3	SDLC Model	5
2.3.1	Why Agile Was Selected	5
2.3.2	Description of Agile Implementation in Digital Market Yard System	5
2.3.3	Summary of Agile Benefits in Digital Market Yard System	6
3	SYSTEM DESIGN	7
3.1	System Architecture	7
3.2	Analysis model	8
3.2.1	ER Diagram	11
3.2.2	DFD Level 0	13
3.2.3	DFD Level 1	13
3.2.4	DFD Level 2	14
3.3	UML Diagrams	15
3.3.1	Use Case Diagram	15
3.3.2	Class Diagram	16
3.3.3	Sequence Diagram	17
4	REQUIREMENT SPECIFICATION	18
4.1	Hardware Requirements	18
4.2	Software Requirements	19
5	IMPLEMENTATION	20
5.1	Technologies used for Implementation	20
5.2	Module 1	21
5.2.1	User Registration and Authentication	21
5.3	Module 2	24
5.4	Module 3	26
6	CONCLUSION AND FUTURE SCOPE	30
A	Appendix : List of Abbreviations	31
B	References	32

List of Figures

2.1	Agile Methodology Diagram	6
3.1	System Architecture	7
3.2	farmer	8
3.3	Merchant	9
3.4	Admin	10
3.5	Farmer entity	11
3.6	Merchant entity	12
3.7	Level 0 DFD Of Digital Market Yard	13
3.8	Level 1 DFD Of Digital Market Yard	13
3.9	Level 2 DFD Of Digital Market Yard	14
3.10	Use Case Diagram	15
3.11	Class Diagram	16
3.12	Sequence Diagram	17
5.1	Farmer and Merchant login	21
5.2	Farmer and Merchant Registration	22
5.3	Dashboard	23
5.4	Home Page	23
5.5	Bidding Process	24
5.6	Add Product	25
5.7	farmer and merchant notification through email	26
5.8	farmer and merchant bill	27
5.9	Management	28
5.10	Product History	29
5.11	Otp Verification	29

List of Tables

4.1	Hardware Requirements for Digital Market Yard System	18
4.2	Software Requirements for Digital Market Yard System	19
5.1	Functional Modules of Digital Market Yard System	28
A.1	List of Abbreviations	31

Chapter 1

INTRODUCTION

The Digital Market Yard System is a modern web-based platform designed to revolutionize the traditional mandi (agricultural market yard) system. In many regions, these market yards still rely on outdated manual processes, such as paper-based record keeping and physical bidding, which lead to inefficiencies, delays, and financial inaccuracies.

Farmers often lack transparency about the prices of their crops and struggle to track their earnings. Merchants face issues managing inventory and transactions, while brokers handle crop weights and payments manually, which slows down the entire system.

This project addresses these challenges by providing a unified digital solution that brings farmers, merchants, and brokers (admins) onto a single platform. The system enables online crop listing, real-time bidding, automated invoicing, and secure transactions — creating a more transparent, efficient, and trustworthy agricultural trading environment.

Built using Spring Boot, Java 17, Thymeleaf, and MySQL, the platform ensures robust backend performance and user-friendly frontend design. The system also integrates Spring Security for authentication and role-based access, and is designed to scale in the future with mobile apps, multilingual support, and analytics dashboards.

1.1 Introduction to Project

The traditional market yard system in agriculture has been in use for a long time, but it is now outdated and inefficient. Several issues plague the current system:

Farmers face challenges in tracking their income, understanding crop pricing, and ensuring fair trade.

Merchants struggle to maintain records of daily purchases and payment histories.

Brokers often record data manually, making it difficult to manage product weights, finances, and timely payments.

These manual processes result in frequent delays, errors, and financial losses. Hence, there is a pressing need for a digital platform that streamlines market yard operations and improves efficiency for all stakeholders.

1.2 Problem Statement

Traditional markets don't use digital tools. Farmers don't know their real earnings after costs. Merchants can't track daily sales. Brokers can't track weights, expenses, or pay farmers quickly. Old methods cause delays, mistakes, and money loss. Hence, we are developing Digital Market Yard, an application designed to streamline transactions, track earnings, and enhance efficiency for farmers, merchants, and brokers.

Thus, to overcome these challenges, we propose "Digital Market Yard", a web application that streamlines agricultural transactions and introduces automation and transparency in the entire trading cycle.

1.3 Scope and Objectives

1.3.1 Scope

The scope of the Digital Market Yard project is to provide a centralized digital platform for all key stakeholders—farmers, merchants, and brokers—to manage market yard operations efficiently. The platform replaces manual, paper-based processes with automated systems for tracking, transactions, and reporting.

Key areas covered under the scope:

1. Farmers can track earnings and expenses digitally.
2. Merchants can manage sales, inventory, and generate analytical reports.
3. Brokers can record crop weights, manage expenses, and process quick payments.
4. The system ensures secure user access, data transparency, and error-free digital records.
5. Supports automation and digitalization of all mandi-level operations.

1.3.2 Objective

Today, farming and agriculture are changing fast, and there is a need for a simple digital system that helps farmers, merchants, and brokers work together easily. Many still use pen and paper to keep records, which can lead to mistakes and delays. This project aims to solve those problems by using a modern app to handle farm transactions, payments, and records in one place. Below are the main goals of this platform:

1. Digitize Agricultural Transactions: Provide a digital platform to automate product listing, inventory management, bidding, order tracking, and invoice generation, reducing dependency on manual systems.
2. Enable Role-Based Access: Empower farmers, merchants, and brokers with customized access to manage crops, procurement, logistics, and payments according to their roles.
3. Ensure Real-Time Monitoring and Reporting: Allow stakeholders to track income, expenses, weight logs, and transactions in real-time, while generating automated monthly and annual reports.
4. Enhance Transparency and Accuracy: Reduce manual errors, delays, and miscommunication by maintaining secure and transparent digital records, improving trust among users.
5. Promote Usability and Inclusivity: Design the platform to be user-friendly and accessible, even for users with limited technical skills, ensuring widespread adoption.

1.4 Identification of Need

1. **Manual Workload is High:** Traditional agricultural markets rely on pen-and-paper systems, which increases the workload and leads to frequent errors.
2. **Lack of Transparency:** Farmers and merchants have no reliable way to verify pricing, transaction status, or commission deductions, causing mistrust.
3. **Delayed Payments:** Brokers manually calculate weights and commissions, which delays payments to farmers and can cause financial stress.
4. **No Centralized Records :** Transaction history, order tracking, and invoice records are scattered or non-existent, making audits or reviews difficult.
5. **Limited Access to Market Data :** Farmers often have no insight into demand, pricing trends, or bidding opportunities, limiting their decision-making power.
6. **Poor Communication :** There's no real-time channel between farmers, merchants, and brokers, leading to confusion and disputes.
7. **No Real-Time Bidding or Order Matching :** Crops are sold via fixed pricing, missing out on potential higher bids due to lack of auction/bidding support.
8. **No Data Security or Backup :** Manual records can be lost or tampered with. A digital system can ensure backup, access control, and integrity.
9. **Scalability is Difficult :** The manual model cannot handle growing numbers of farmers, crops, and transactions in a scalable way.
10. **Inaccessibility of Existing Government Systems :** Platforms like eNAM are either too complex or underused in many rural regions due to lack of awareness or infrastructure.

Chapter 2

LITERATURE REVIEW

The Indian agricultural market still depends heavily on traditional mandi (market yard) systems, which are often unorganized and inefficient. Farmers, merchants, and brokers rely on manual processes to manage transactions, track sales, and maintain records. These outdated methods result in numerous problems such as delays, mismanagement, data inaccuracy, and lack of transparency.

There is growing global and national interest in using digital solutions to modernize agricultural systems. Digital platforms can improve accessibility, streamline operations, and enhance financial transparency. Reports by the Food and Agriculture Organization (FAO) and Government of India stress the importance of digitization in agricultural marketing, and initiatives like eNAM (Electronic National Agriculture Market) reflect this need.

However, many farmers in rural areas still face barriers such as lack of awareness, limited access to digital infrastructure, and dependency on intermediaries. This literature review explores current systems, identifies their limitations, and highlights the gaps our proposed system aims to fill.

2.1 Literature Review

Several academic and governmental studies have stressed the importance of digital solutions in agricultural markets. Platforms like eNAM (Electronic National Agriculture Market) show promise but often face challenges in adoption and integration. Research shows that most farmers are not well-versed in using complex digital systems and need platforms that are intuitive and require minimal training.

Furthermore, literature indicates a strong need for features such as real-time tracking, automated billing, inventory management, and direct merchant access to truly empower farmers.

2.2 Existing Systems

A number of digital platforms currently operate in this space:

eNAM: A government platform designed to unify agricultural markets across India, but adoption remains limited to select regions.

AgriBazaar: A private digital marketplace for crops with some level of bidding and online payment support.

Agmarknet: Focuses on agricultural price dissemination but lacks transaction capabilities.

Local Mandi Apps: Some state governments have launched regional apps for farmer registration and price tracking, but most lack robust backend infrastructure or scalability.

These systems, while effective to a degree, generally lack full integration of automated bidding, digital invoicing, cloud data storage, and multi-stakeholder roles which are essential for end-to-end digitization.

2.3 SDLC Model

Agile is a modern software development methodology that emphasizes adaptability, collaboration, and delivering functional software in small, frequent increments. Rather than building the entire system at once, Agile divides the project into smaller components known as iterations or sprints. Each sprint results in a usable module that can be tested and refined based on stakeholder feedback. This approach fosters constant communication between the development team and end users, ensuring the solution evolves to meet real-world needs and changing requirements.

2.3.1 Why Agile Was Selected

Agile was selected for the development of the **Digital Market Yard System** because of its ability to align with the dynamic requirements of a web-based agricultural marketplace. Key reasons include:

- **Flexibility:** Agile allows the team to adapt quickly to changing agricultural policies, stakeholder inputs, or technology trends.
- **User Feedback:** Regular testing and feedback from farmers, traders, and admin users help enhance usability and fix issues early.
- **Incremental Delivery:** Essential modules like registration, product listing, bidding, and payment can be developed and released in stages.
- **Faster Prototyping:** Technologies like Spring Boot and MySQL support rapid development cycles and integration.
- **Scalability:** Agile accommodates future additions such as multilingual support, AI-based price suggestions, and integration with e-Government services.

2.3.2 Description of Agile Implementation in Digital Market Yard System

The **Digital Market Yard System** project was executed using Agile by dividing the development into multiple sprints. Each sprint focused on core features such as:

- User registration and authentication
- Crop listing and real-time bidding
- Transaction and payment management
- Invoice generation and reporting

After each sprint, the implemented features were tested, and feedback was collected from target users like farmers, merchants, and brokers. This cycle of development, testing, and feedback enabled early detection of bugs, continuous refinement of the system, and an improved user experience.

Using Spring Boot and MySQL for backend and Thymeleaf for UI ensured real-time data handling and rapid deployment. Agile's iterative nature allowed modular development, enabling the team to enhance the system progressively with minimal disruptions.

2.3.3 Summary of Agile Benefits in Digital Market Yard System

- Feature development in smaller, manageable units.
- Continuous user testing and feedback integration.
- Quick adaptation to evolving requirements.
- Fast and effective prototyping using modern tools.
- Easy integration of future features and scalability.

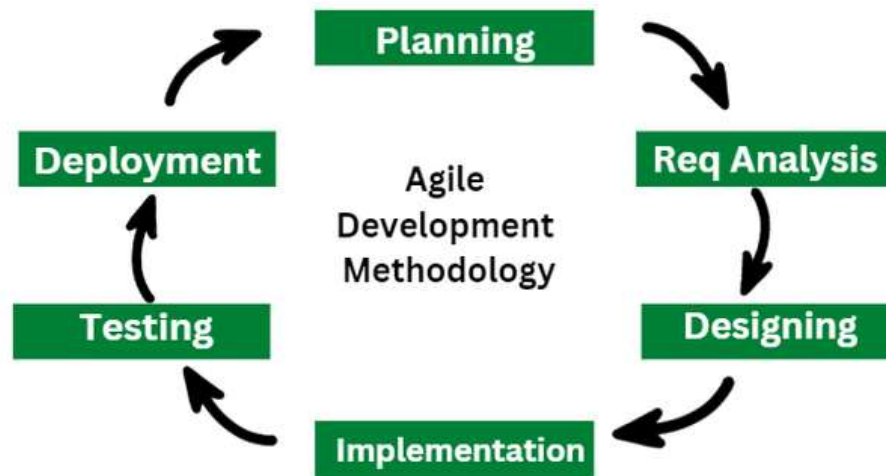


Figure 2.1: Agile Methodology Diagram

Chapter 3

SYSTEM DESIGN

3.1 System Architecture

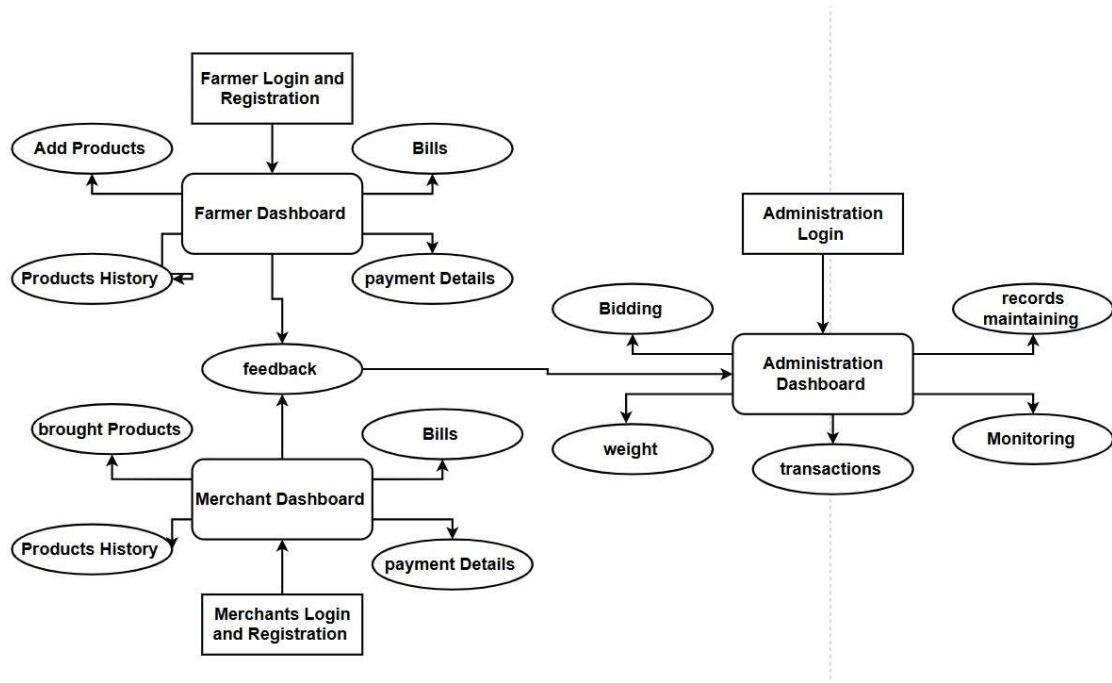


Figure 3.1: System Architecture

3.2 Analysis model

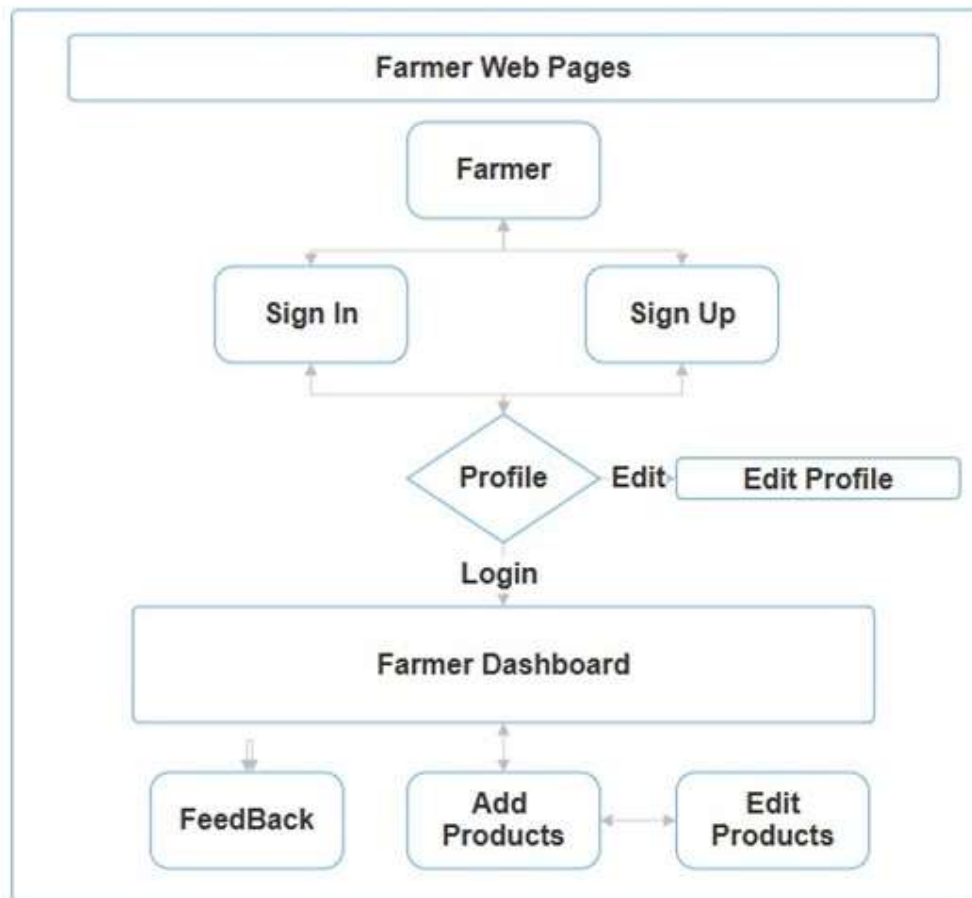


Figure 3.2: farmer

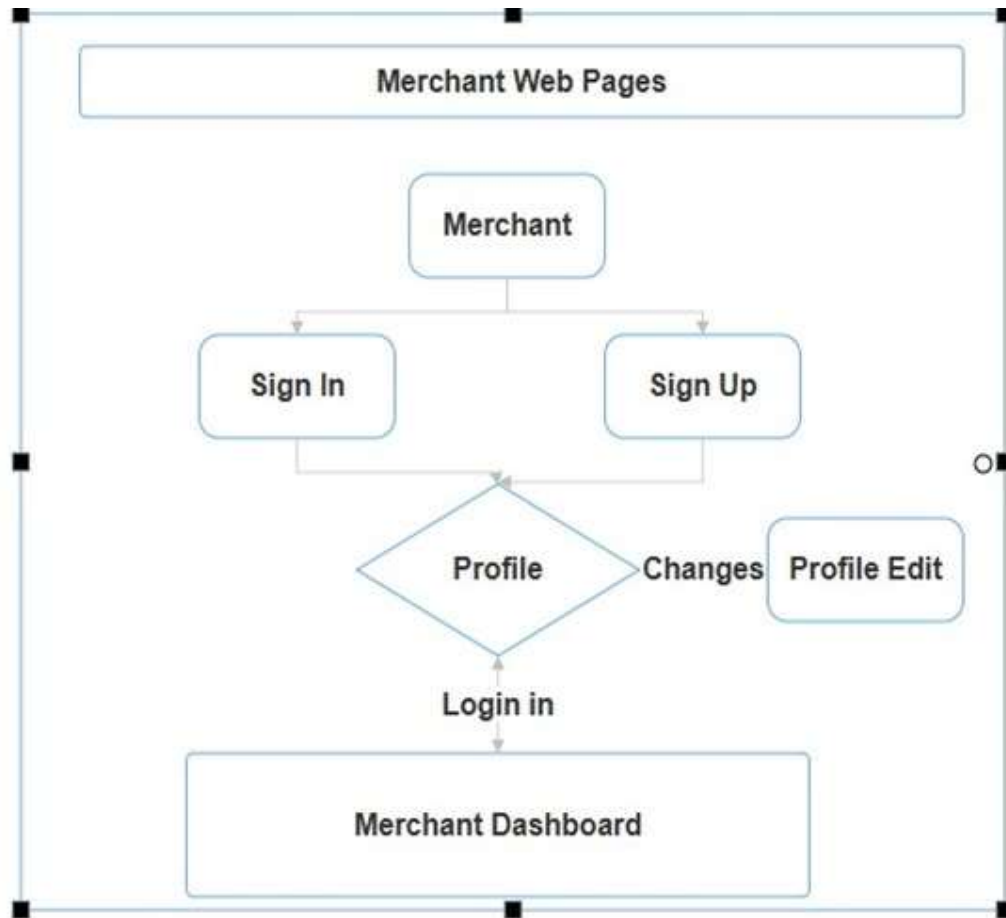


Figure 3.3: Merchant

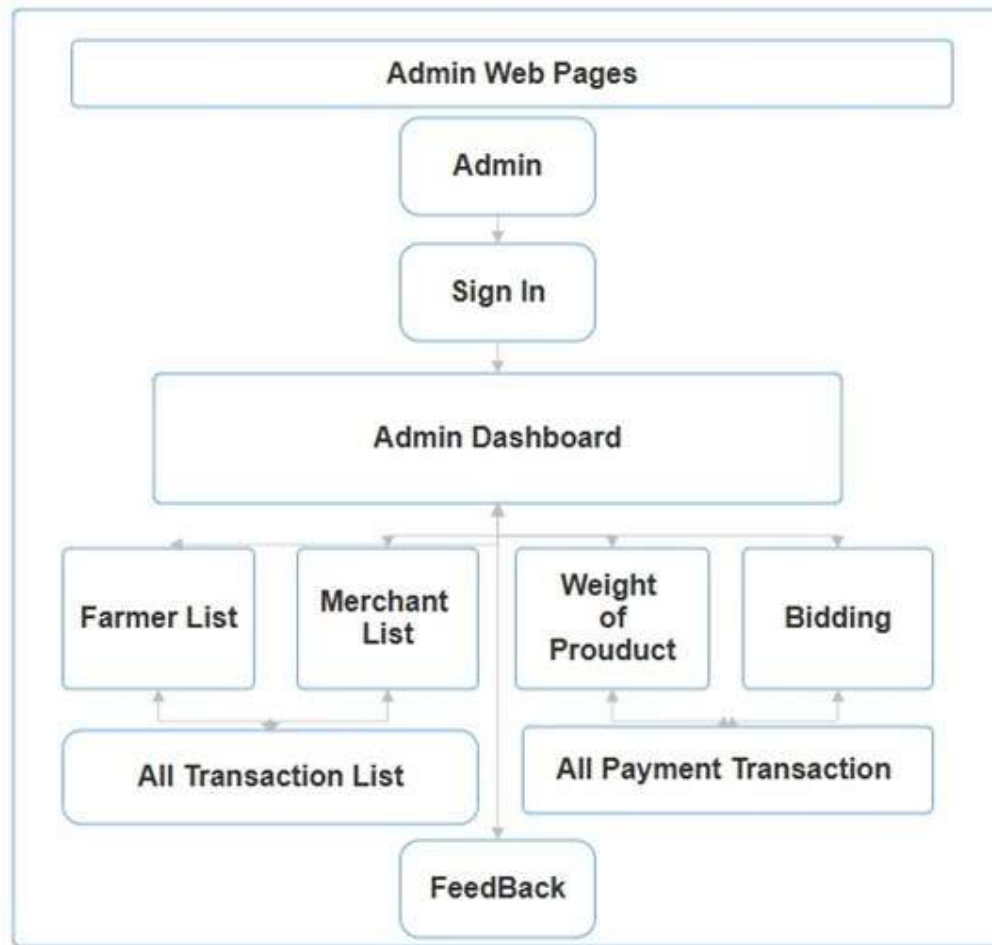


Figure 3.4: Admin

3.2.1 ER Diagram

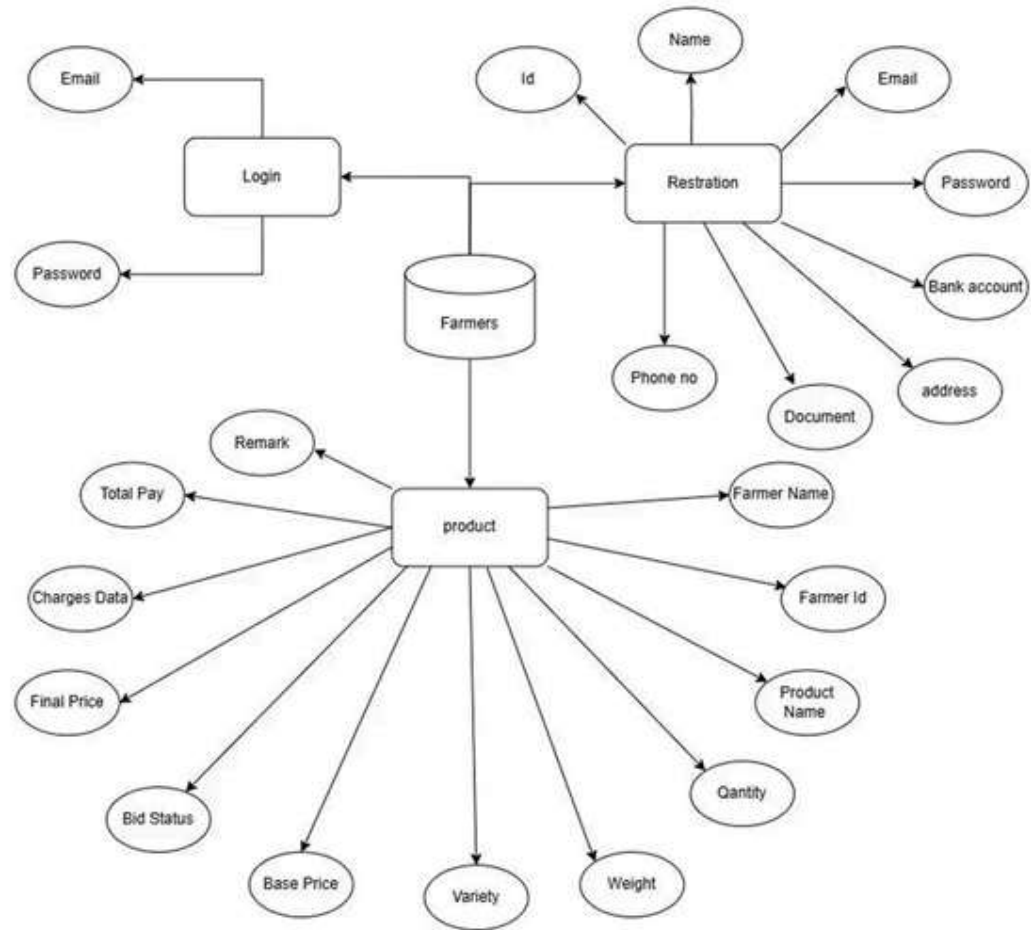


Figure 3.5: Farmer entity

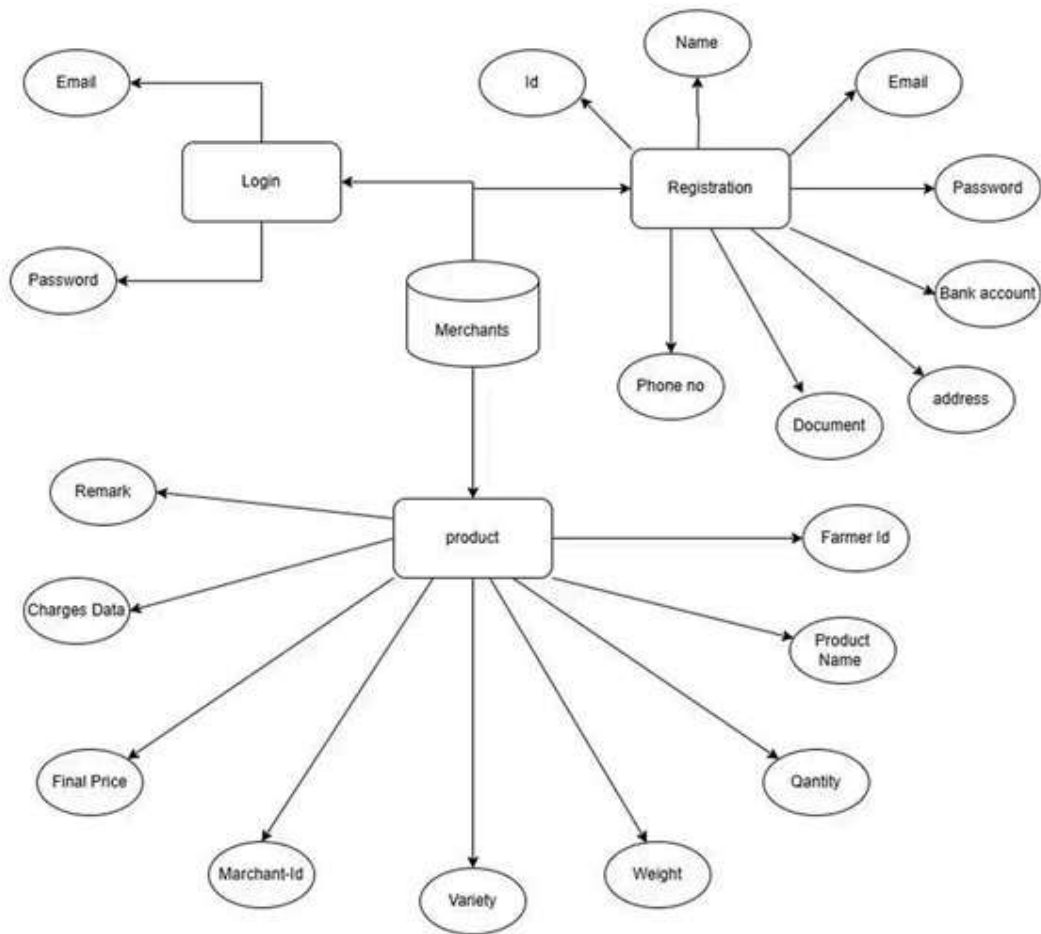


Figure 3.6: Merchant entity

3.2.2 DFD Level 0

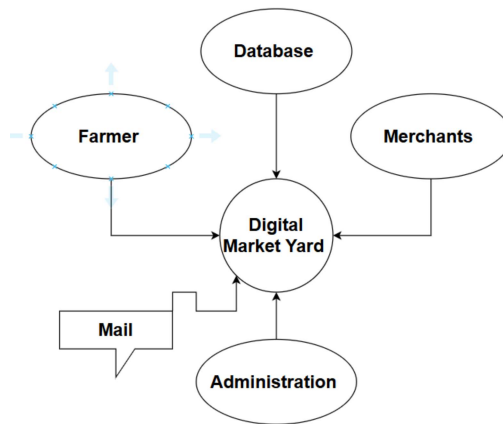


Figure 3.7: Level 0 DFD Of Digital Market Yard

3.2.3 DFD Level 1

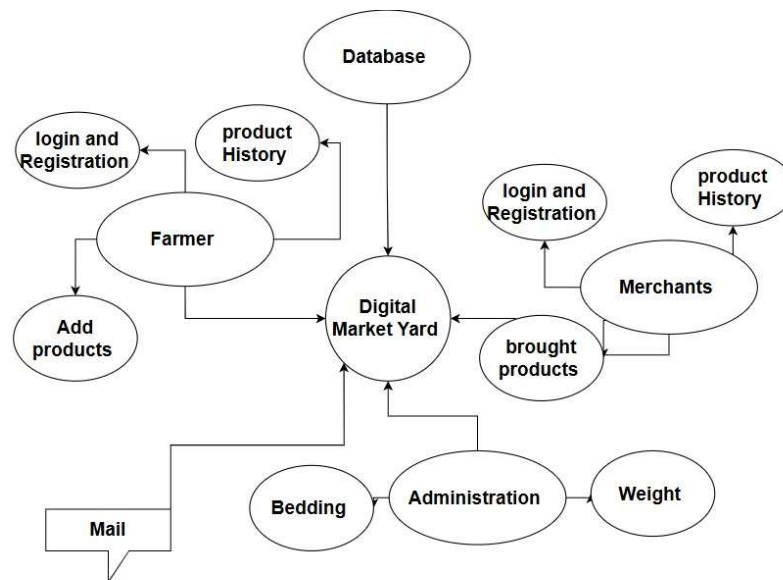


Figure 3.8: Level 1 DFD Of Digital Market Yard

3.2.4 DFD Level 2

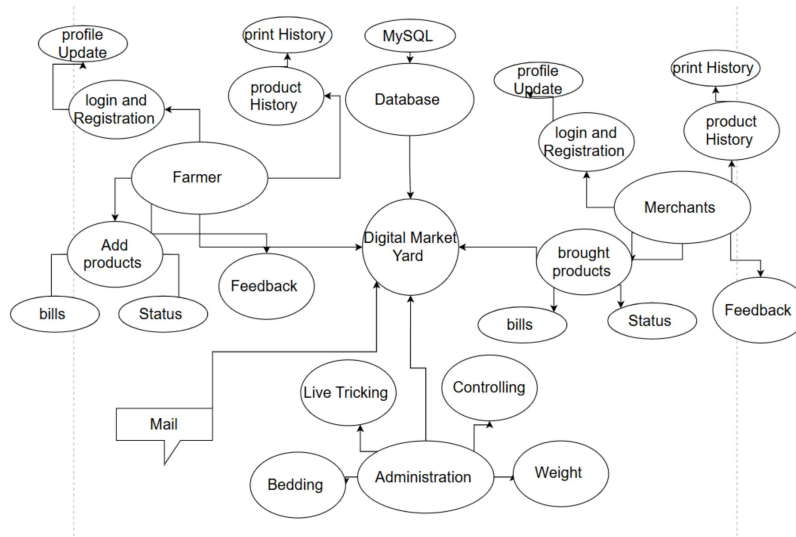


Figure 3.9: Level 2 DFD Of Digital Market Yard

3.3 UML Diagrams

3.3.1 Use Case Diagram

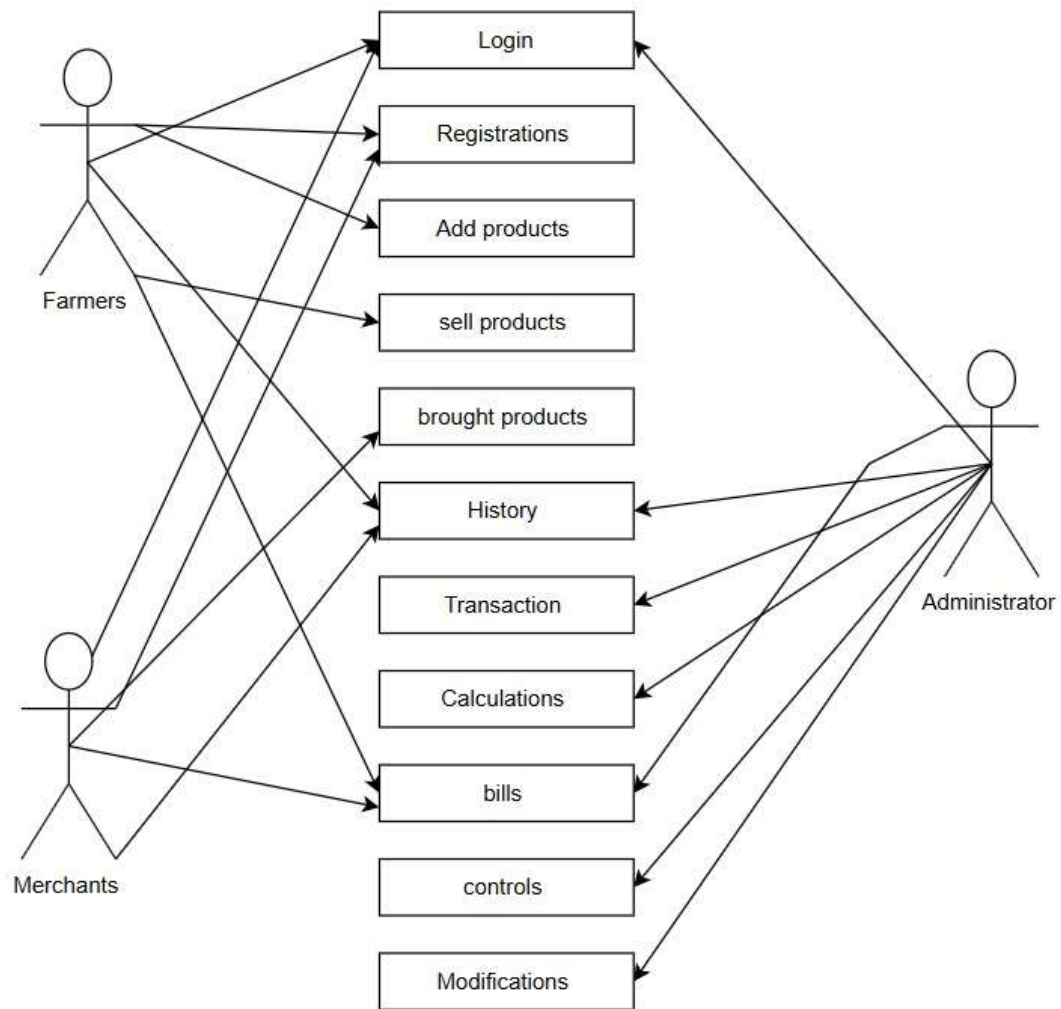


Figure 3.10: Use Case Diagram

3.3.2 Class Diagram

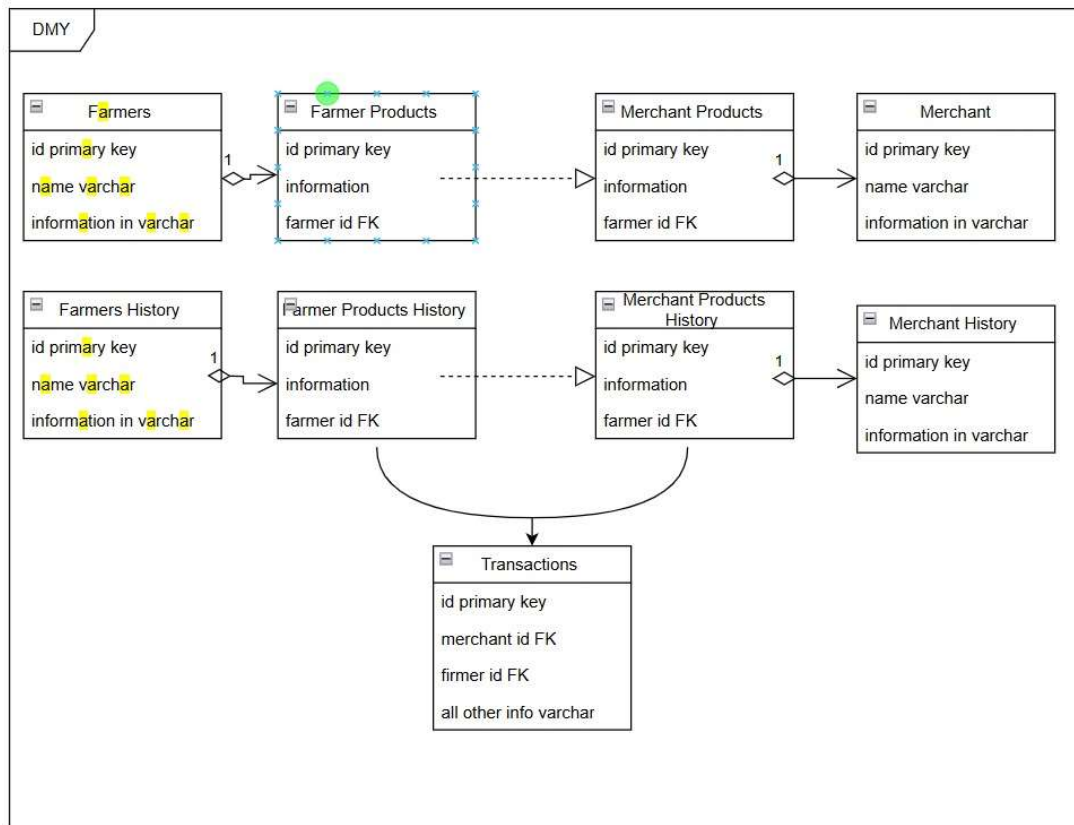


Figure 3.11: Class Diagram

3.3.3 Sequence Diagram

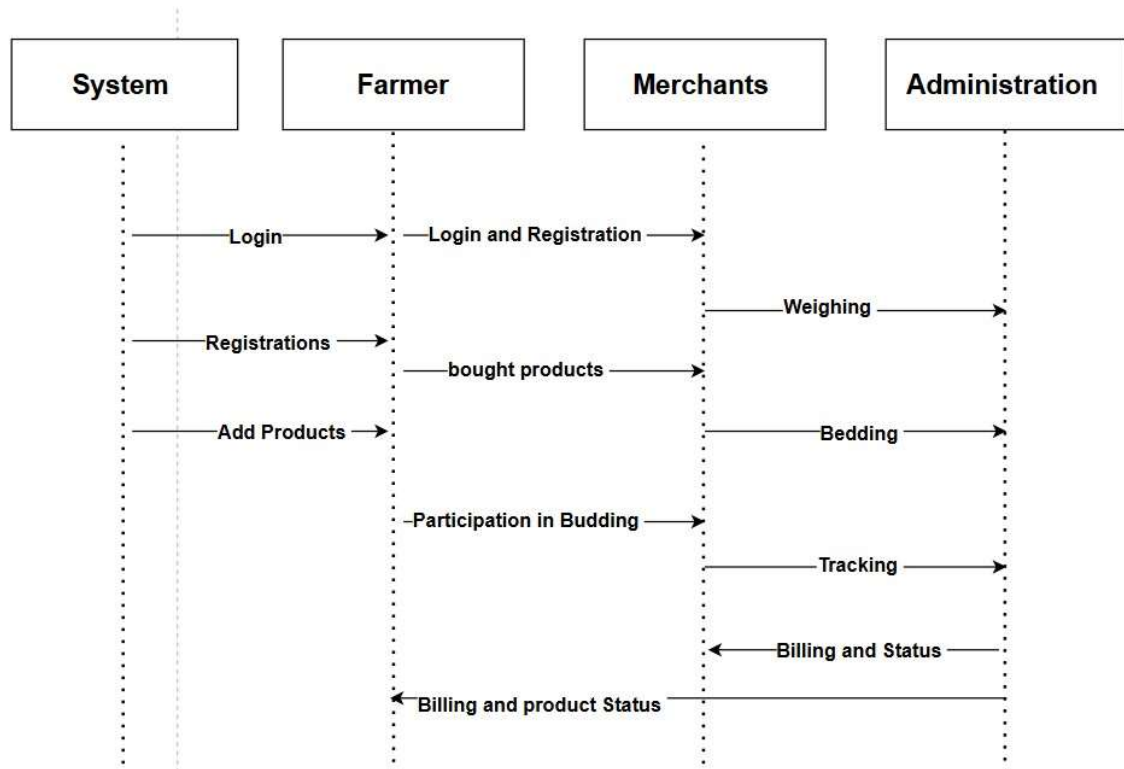


Figure 3.12: Sequence Diagram

Chapter 4

REQUIREMENT SPECIFICATION

4.1 Hardware Requirements

To develop and deploy the Digital Market Yard web application effectively, the following hardware configuration is recommended:

- Processor: Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent
- RAM: Minimum 8 GB (Recommended: 16 GB for smoother development experience)
- Storage: 256 GB SSD or higher (for fast read/write operations)
- Display: Full HD (1920x1080) resolution for better UI design
- Internet: Stable broadband connection (for Firebase hosting and online dependencies)

Component	Recommended Specification
Processor	Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent
RAM	Minimum 8 GB (Recommended: 16 GB)
Storage	256 GB SSD or higher
Display	Full HD (1920x1080) resolution
Internet	Stable broadband connection (for Firebase and online dependencies)

Table 4.1: Hardware Requirements for Digital Market Yard System

4.2 Software Requirements

- Operating System: Windows 10 / Linux (Ubuntu) / macOS
- Programming Language: Java 17
- Backend Framework: Spring Boot (MVC architecture)
- Frontend Technologies: HTML, CSS, JavaScript, Thymeleaf
- Database: MySQL (current) and Firebase (for future cloud deployment)
- Build Tool: Maven
- Authentication Framework: Spring Security
- IDE: IntelliJ IDEA / Eclipse
- Version Control: Git and GitHub
- Browser: Google Chrome / Mozilla Firefox (for testing UI)

Component	Specification
Operating System	Windows 10 / Linux (Ubuntu) / macOS
Programming Language	Java 17
Backend Framework	Spring Boot (MVC architecture)
Frontend Technologies	HTML, CSS, JavaScript, Thymeleaf
Database	MySQL (current), Firebase (future cloud deployment)
Build Tool	Maven
Authentication Framework	Spring Security
IDE	IntelliJ IDEA / Eclipse

Table 4.2: Software Requirements for Digital Market Yard System

Chapter 5

IMPLEMENTATION

5.1 Technologies used for Implementation

- **Frontend:** HTML, CSS, and JavaScript are used to design responsive user interfaces for farmers, merchants, and brokers.
- **UI Engine:** Thymeleaf is integrated with Spring Boot to generate dynamic content on the server side.
- **Backend:** Spring Boot (Java 17) is used to develop RESTful APIs, manage business logic, and connect to databases securely and efficiently.
- **Database:** MySQL is the primary relational database used for storing user data, crops, transactions, and reports. Firebase Firestore is planned for cloud-based scalability.
- **Authentication:** Spring Security ensures secure login, registration, and role-based access control for all system users.
- **Build Tool:** Maven manages project dependencies, builds, and packaging of the Java application.
- **Hosting:** Firebase Hosting (planned) or other platforms like AWS/Heroku may be used to deploy the web application.
- **IDE:** IntelliJ IDEA and Eclipse are the primary development environments for writing, testing, and debugging the application.
- **Version Control:** Git and GitHub are used for source code management, collaboration, and version tracking.
- **Diagram Tools:** draw.io, Lucidchart, and PlantUML are used to design ER diagrams, DFDs, UML use case diagrams, and class/sequence diagrams.
- **Testing Tools:** Google Chrome and Mozilla Firefox are used for frontend UI testing. Postman is used to test APIs, and JUnit is used for unit testing backend functionalities.

5.2 Module 1

5.2.1 User Registration and Authentication

Frontend Implementation (HTML, CSS, Thymeleaf):

- Login and Registration Pages are created using HTML/CSS with Thymeleaf integration for dynamic rendering.
- Forms include:
 - Name
 - Email
 - Phone
 - Password
 - Document
 - Address
- Form validation is performed server-side using Spring Boot.

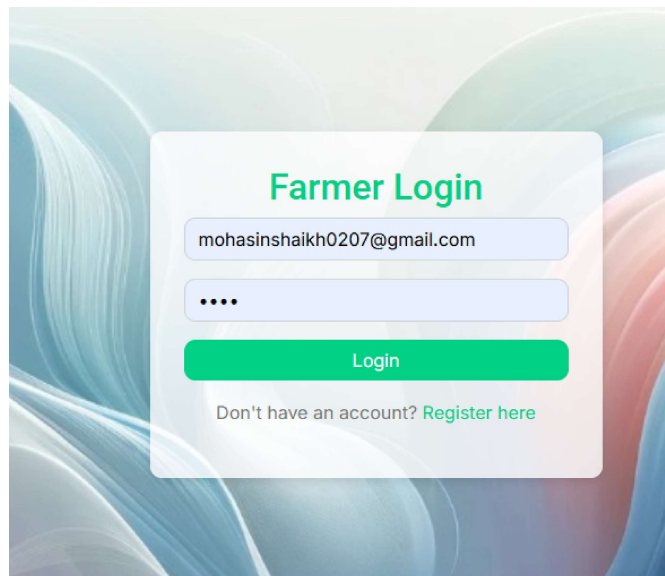


Figure 5.1: Farmer and Merchant login

Backend Implementation (Spring Boot, Java 17):

- Developed using Spring Boot MVC architecture.
- Controllers handle HTTP requests such as `/login`, `/register`.
- Services implement business logic like user saving, role validation, and password encoding.

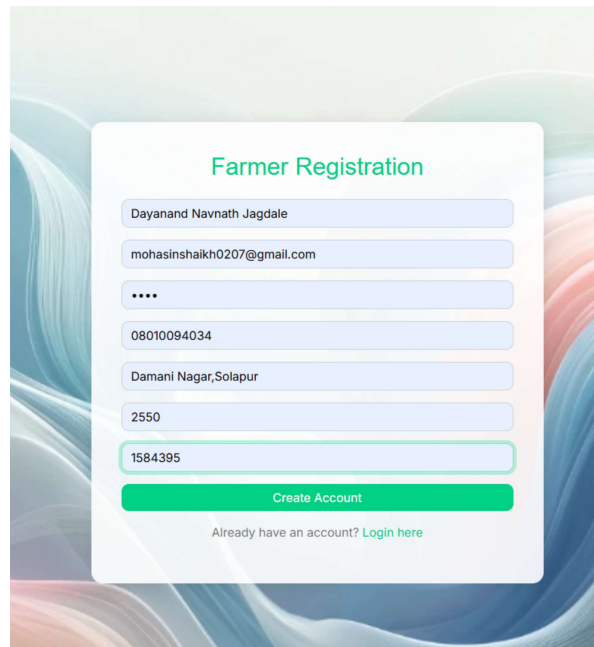
A screenshot of a 'Farmer Registration' form. The form is titled 'Farmer Registration' in green text. It contains several input fields with the following values: Name: 'Dayanand Navnath Jagdale', Email: 'mohasinshaikh0207@gmail.com', Password: '****', Phone: '08010094034', Address: 'Damani Nagar, Solapur', Pin: '2550', and Aadhar: '1584395'. Below the fields is a green 'Create Account' button. At the bottom, there is a link: 'Already have an account? [Login here](#)'.

Figure 5.2: Farmer and Merchant Registration

- **Spring Security** handles:
 - User authentication with email verification.
 - Role-based authorization (Farmer, Merchant, Admin access).
 - Session management (login/logout).
- **Database (MySQL)** stores user information in a table named **users**, which includes:
 - **id** – Primary Key
 - **name**
 - **email**
 - **password**
 - **document**
 - **address**

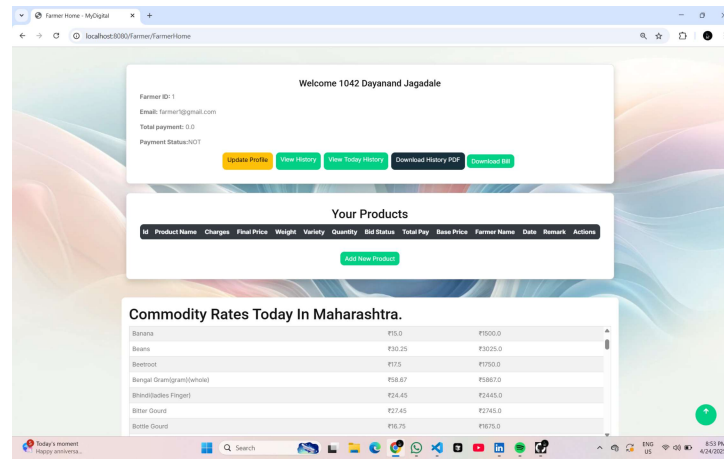


Figure 5.3: Dashboard

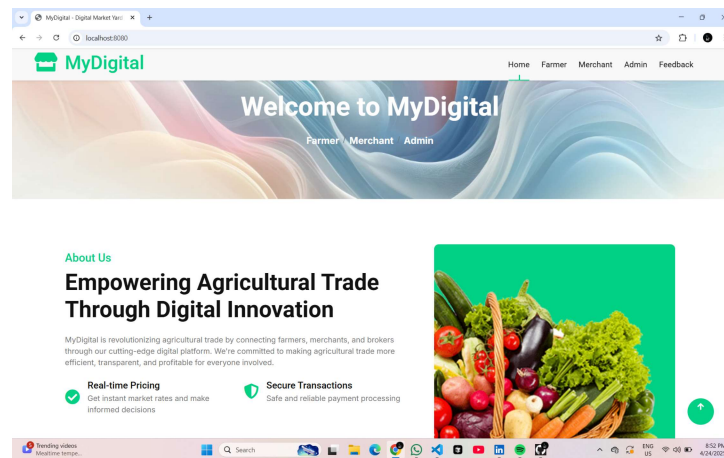


Figure 5.4: Home Page

5.3 Module 2

Crop Listing and Real-time Bidding

This module forms the core functionality of the Digital Market Yard System, enabling dynamic interaction between farmers and merchants in a transparent, competitive, and efficient environment. It replaces traditional crop selling methods with a digital process that supports both fixed-price transactions and real-time bidding.

- **Crop Listing by Farmers:** Farmers can create listings for the crops they intend to sell, including details such as crop name, quantity, expected price, quality grade, and optional images or descriptions.
- **Direct Orders and Bidding by Merchants:** Merchants can browse the listings and either place a direct order at the listed price or participate in live bidding. Bids are stored in real-time, and merchants can see competing offers to encourage fair pricing.
- **Real-time Auction Handling:** The system facilitates live bidding sessions with automated timeouts. Bids are ranked, and the highest bid is highlighted for easy tracking. Farmers receive instant notifications of new bids.

The screenshot displays the Admin Dashboard interface. On the left is a sidebar with navigation links: Admin, Dashboard, Transactions, Farmers, Merchants, Farmer Products, Merchant Product, Weight, and Bidding (highlighted). The main content area is titled 'Product Details' and contains the following information:

Product ID: 1	Variety: kokan
Product Name: mango	Quantity: 2000
Bidding Status: NOT	Date: 2025-04-24
Base Price: 200000.0	Weight: 0.0

Below this is the 'Farmer Details' section:

Farmer Name: 1042 Dayanand Jagadale	Phone: 08010064034
Email: farmer1@gmail.com	Address: Damani Nagar

At the bottom is the 'Save Product to Merchant' section, which includes a form with 'Enter Merchant ID' (value: 1) and 'finalPrice' (value: 500000), a 'Save to Merchant' button, and 'Previous' and 'Next' navigation buttons.

Figure 5.5: Bidding Process

- **Admin/Broker Supervision:** The Admin (Broker) oversees the entire bidding and transaction process. They can resolve disputes, moderate listings, and ensure all activities are fair and within system guidelines.
- **Automatic Fee and Tax Calculation:** The platform automatically calculates broker commissions, taxes, and other charges before finalizing the transaction. This ensures transparency and reduces manual errors.

- **Transaction History and Transparency:** Every crop listing, bid, and transaction is digitally stored. This enables users to view history, generate reports, and verify records at any time, fostering trust among all parties.
- **Benefits:** This module promotes transparency, eliminates middlemen exploitation, supports better pricing through competition, and improves overall market efficiency for both farmers and merchants.

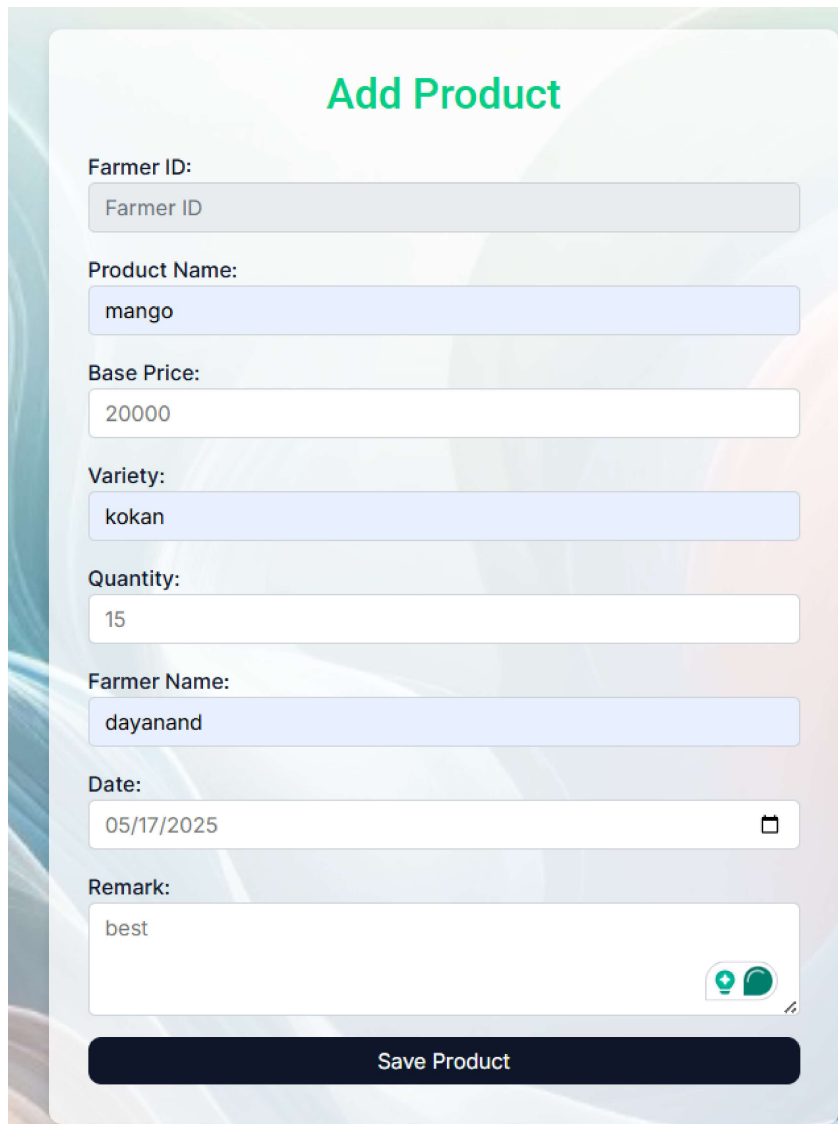
A screenshot of a web application form titled "Add Product" in green text. The form is set against a light blue and green abstract background. It contains several input fields with labels: "Farmer ID:" with a grey input box containing "Farmer ID"; "Product Name:" with a blue input box containing "mango"; "Base Price:" with a white input box containing "20000"; "Variety:" with a blue input box containing "kokan"; "Quantity:" with a white input box containing "15"; "Farmer Name:" with a blue input box containing "dayanand"; "Date:" with a white input box containing "05/17/2025" and a calendar icon; and "Remark:" with a white text area containing "best". At the bottom right of the text area are two circular icons, one green with a plus sign and one dark green. Below the form is a dark blue button with the text "Save Product" in white.

Figure 5.6: Add Product

5.4 Module 3

Transaction Management and Reporting

This module is essential for ensuring transparency, accountability, and operational efficiency within the Digital Market Yard System. It handles the complete post-sale process, including order tracking, payment verification, invoice generation, and analytics reporting. The system digitizes these processes to eliminate manual paperwork, improve accuracy, and build trust among all stakeholders.

- **Order and Payment Tracking:** Farmers and merchants can monitor the status of their orders in real time, including confirmation, dispatch, and delivery. Payment statuses—such as pending, processed, or completed—are also displayed transparently in user dashboards.
- **Weight Verification and Payment Approval:** Admins (Brokers) are responsible for verifying crop weights and confirming product quality at the time of dispatch or delivery. Once verified, payments are approved and disbursed to the farmer. This ensures fairness and accuracy in every transaction.

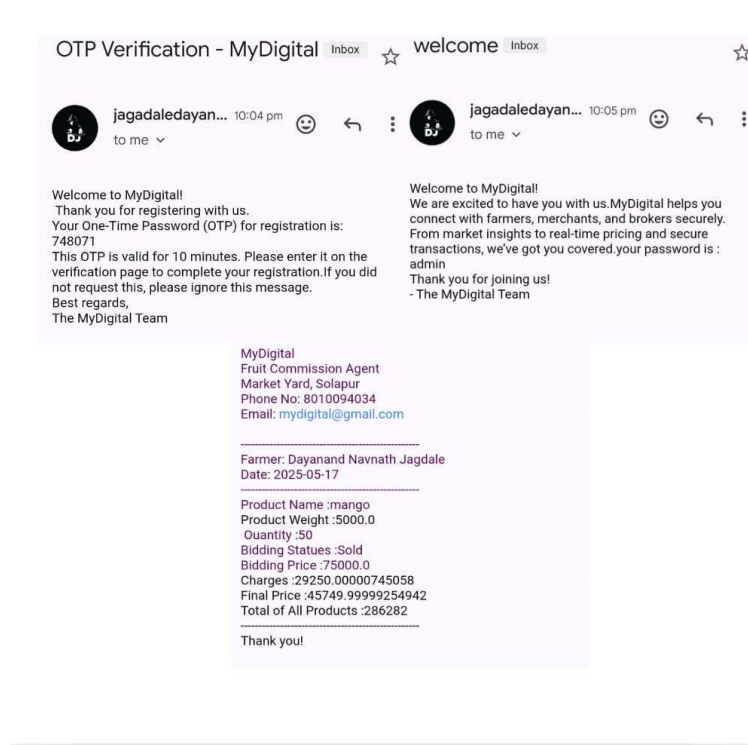


Figure 5.7: farmer and merchant notification through email

- **Automatic Invoice Generation:** After transaction confirmation, a digital invoice is automatically generated. This invoice includes:
 - Crop name, quantity, and price
 - Farmer and merchant details

- Taxes, commissions, and total payable amount

All invoices are securely stored and accessible to both parties.

MyDigital					
Fruit Commission Agent					
Market Yard, Solapur					
Phone No: 8010094034					
Email: mydigital@gmail.com					

Farmer ID: 4					
Farmer: Dayanand Navnath Jagdale					
Farmer Phone No: 08010094034					
Date: 2025-05-17					

Qty	Product	Weight	Charges (₹)	Price (₹)	Total (₹)
1	mango	150.00	877.50	10,000.00	9,122.50
1	mango	250.00	1,462.50	10,000.00	8,537.50
1	kanda	0.00	0.00	0.00	0.00

Total : 17,660.00					

Thank you!					

Figure 5.8: farmer and merchant bill

- **Reporting and Analytics:** The system provides detailed monthly reports and dashboards that include:
 - Total sales and purchases
 - Most frequently sold crops
 - Pending and completed transactions
 - Income and expenditure summaries

These insights help stakeholders make informed decisions.

- **Paperless and Accurate Documentation:** By automating invoice and report generation, the system minimizes human error and avoids the need for physical record-keeping. This enhances reliability and makes the system audit-ready.
- **Improved Trust and Transparency:** With all transactions digitally verified, and records securely stored, all users — farmers, merchants, and admins — benefit from increased trust, reduced disputes, and faster payment cycles.

Module	Description
User Registration and Authentication	- Farmers, Merchants, and Admins (Brokers) can register and log in securely. - Role-based access control restricts access to respective user functions. - Passwords are encrypted, and sessions are securely managed using Spring Security. - Users can update their profiles and reset passwords.
Crop Listing and Real-time Bidding	- Farmers can list crops with details such as quantity, price, and description. - Merchants can browse, place direct orders, or bid in real time. - Admins oversee bidding and resolve disputes. - The system auto-calculates fees and taxes. - All bids and transactions are stored for transparency.
Transaction Management and Reporting	- Users can track order statuses, payments, and deliveries. - Admins verify weights and approve payments. - Invoices with tax and commission breakdowns are generated. - Monthly analytics and reports are available to all users. - Reduces paperwork and builds trust through automation.

Table 5.1: Functional Modules of Digital Market Yard System

Farmers Management

Total Farmers

5

Farmer List

+ Add New Farmer

ID	Name	Email	Phone	Address	Account No	Document	Actions
1	farmer1	farmer1@gmail.com	08010094034	Damani Nagar	1584395	2550	
2	farmer 2	farmer2@gmail.com	08010094034	Damani Nagar	1584395	2550	
3	farmer3	farmer3@gmail.com	08010094034	Damani Nagar	1584395	2550	
4	Dayanand Navnath Jagdale	rohithbandgar667@gmail.com	08010094034	Damani Nagar, Solapur	1584395	10	
5	Saurabh Aglave	admin@gmail.com	08010094034	Damani Nagar	1584395	2550	

Figure 5.9: Management

Product History Report

Product Name	Charges	Final Price	Weight	Quantity	Bid Status	Total Pay	Date
mango	877.50000 02235174	10000.0	150.0	150	SOLD	9122.4999 99776483	2025-05-09
mango	1462.5000 00372529	10000.0	250.0	98	SOLD	8537.4999 99627471	2025-05-09

Figure 5.10: Product History

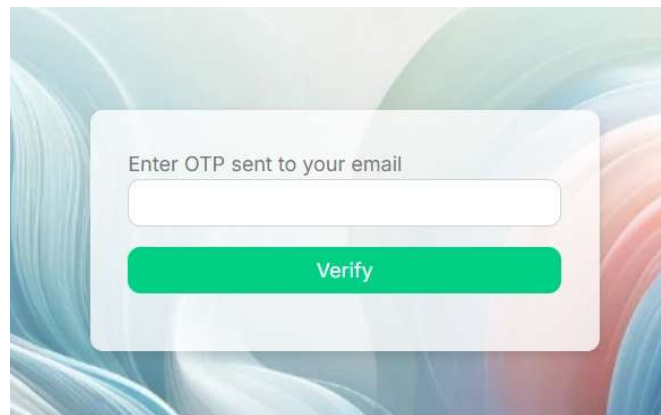
A screenshot of an OTP verification form. The form is a light gray rounded rectangle with a white border, centered on a background with abstract blue and green wavy patterns. Inside the form, the text "Enter OTP sent to your email" is displayed in a small, gray font. Below this text is a white rectangular input field with a thin gray border. At the bottom of the form is a solid green rectangular button with the word "Verify" written in white text.

Figure 5.11: Otp Verification

Chapter 6

CONCLUSION AND FUTURE SCOPE

Conclusion

The Digital Market Yard System effectively bridges the gap between traditional agricultural market yards and modern digital technology. By digitizing crop listing, bidding, and transaction processes, the platform empowers farmers, merchants, and brokers (admins) to operate with greater transparency, efficiency, and accountability.

Farmers gain real-time insights into pricing and income, merchants can track inventory and payments accurately, and brokers benefit from automated management of weights, expenses, and payments. The system's automated tax and commission calculations minimize errors and speed up the entire trading cycle.

Overall, this platform enhances trust among stakeholders, reduces dependency on manual paperwork, and streamlines agricultural commerce, paving the way for a more sustainable and profitable farming ecosystem.

Future Scope

- **Mobile Application Integration:** Develop Android and iOS apps to provide farmers and merchants with on-the-go access to the marketplace.
- **Payment Gateway Integration:** Implement secure, seamless online payments for faster and transparent financial transactions.
- **Advanced Search and Filtering:** Improve the user experience by enabling sophisticated crop searches and filtering options.
- **Real-time Chat Functionality:** Facilitate direct communication between farmers and merchants to negotiate and clarify transaction details instantly.
- **Interactive Analytics Dashboard:** Provide comprehensive data visualization tools to help users analyze market trends and performance.
- **Multi-language Support:** Localize the platform for regional languages to make it accessible to a broader base of farmers.
- **Integration with Weather and Crop Advisory Services:** Offer real-time weather updates and crop management suggestions to assist farmers in decision-making.
- **Blockchain-based Transaction Security:** Enhance trust and transparency by incorporating blockchain technology for tamper-proof transaction records.

Appendix A

Appendix : List of Abbreviations

ABBREVIATIONS	ILLUSTRATION
DMY	Digital Market Yard
SDLC	Software Development Life Cycle
ER	Entity Relationship
DFD	Data Flow Diagram
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
MVC	Model View Controller
RDBMS	Relational Database Management System
CRUD	Create, Read, Update, Delete
HTTP	HyperText Transfer Protocol
PDF	Portable Document Format
OTP	One Time Password
MySQL	My Structured Query Language
Thymeleaf	Server-side Java template engine
NBNSCOE	N. B. Navale Sinhgad College of Engineering

Table A.1: List of Abbreviations

Appendix B

References

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- **Digital Market:** Government initiatives on (Electronic National Agriculture Market) – <https://enam.gov.in>

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